

Bira Odra

AI / Machine Learning | Computer Vision | Software Engineering
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Education

University at Buffalo, SUNY

M.S., Computer Science; Artificial Intelligence, Machine Learning & Computer Vision

• **GPA:** 3.83/4.00

• **Coursework:** Deep Learning, Introduction to Machine Learning, Statistical Data Mining II, Algorithms Analysis & Design, Data-Intensive Computing, Data Models & Query Languages, Operating Systems, Computer Security

Aug 2025 - Present

Buffalo, NY

Keshav Memorial Institute of Technology

B.Tech, Computer Science (Artificial Intelligence & Machine Learning); Computer Vision, Deep Learning & NLP

• **GPA:** 8.4/10

• **Coursework:** Deep Learning for Vision Systems, Artificial Intelligence, Introduction to Machine Learning, NLP Using Transformers, Big Data with Hadoop & Spark, Data Structures & Algorithms, Database Management Systems

May 2025

Hyderabad, India

Technical Skills

Languages: Python, C++, Java, C, JavaScript, SQL

AI/ML: Deep Learning, Machine Learning, Computer Vision, NLP, Generative AI, Reinforcement Learning, Statistical Modeling

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, scikit-learn, OpenCV, Pandas

Engineering & Data: Docker, Git, React.js, Node.js, REST APIs, Streamlit, Gradio, MongoDB, Hadoop, Spark, Postman, relational databases

Experience

Risposta Software India Pvt. Ltd.

Software Developer Intern

May 2024 - August 2024

• Engineered Python and JavaScript web applications and REST APIs, optimizing SQL queries to improve page-load time by 30% and reduce data latency by 20%.

• Delivered production-ready features with React.js, Node.js, and relational databases while partnering across teams to diagnose reliability issues.

Accentiq Systems Pvt. Ltd.

Full Stack Developer

June 2025 - August 2025

• Built accessible React components and backend microservices with REST APIs for enterprise workflows.

• Designed relational schemas and optimized CRUD operations, building scalable applications by integrating frontend, backend, and database layers.

Projects

Graduate Researcher

University at Buffalo

Sep 2026 - Present

Buffalo, NY

• **SENTINEL:** Contributing to diffusion-based robot trajectory planning with LiDAR/environmental context representations, Control Barrier Function (CBF)-guided safety constraints, and uncertainty-aware evaluation.

• **EARTH:** Contributing to autonomous excavation research across perception, state estimation, realistic simulation, and safe motion planning for large hydraulic excavators.

Shadow Removal from Real-World Images

PyTorch | U-Net | OpenCV | ISTD | Gradio

• Engineered U-Net image-restoration pipelines in PyTorch and benchmarked **540 ISTD test triplets**, achieving **30.19 dB PSNR** and **0.9514 SSIM**; surpassed the RGB baseline by **3.01 dB PSNR**.

• Implemented image/mask/ground-truth evaluation, checkpoint comparison, and failure-case diagnostics; improved shadow-region PSNR by **2.21 dB** over baseline and built a Gradio interface for interactive inference.

Real-Time Pose Detection

Python | MediaPipe | OpenCV | NumPy

• Implemented webcam pose estimation and a **2-state** curl counter; computed elbow angles from **3 joint landmarks** using NumPy, with **>160°** extension and **<30°** flexion thresholds, live skeleton overlays, and repetition tracking.

Automated Fruit Health Assessment

TensorFlow | MobileNetV2 | Node.js | Streamlit

• Fine-tuned MobileNetV2 with transfer learning and image augmentation on a **13,599-image** fruit-quality dataset spanning **6 classes**; achieved **97% validation accuracy** and a **96.8% F1-score**.

• Outperformed SVM (**85% accuracy**) and Random Forest (**88%**); integrated model inference with a Node.js backend and Streamlit interface.

SkinGPT - Skincare RAG

BM25 | Vector Search | Embeddings | Cross-Encoder Reranking

• Engineered a retrieval-augmented generation pipeline spanning document ingestion, chunking, embedding-based indexing, hybrid BM25/vector retrieval, cross-encoder reranking, and source-cited answer generation.

• Benchmarked retrieval on **1,000 skincare questions**, achieving approximately **75% retrieval precision**.

Publication

Odra Bira, Sreeja Sunkeswaram, Naresh R. Automated Assessment of Fruit Quality Using Transfer Learning and MobileNetV2. IJRASET, 13(4), 166-172, April 2025. DOI: 10.22214/ijraset.2025.68187

Certifications

• Building with the Claude API: (Anthropic)

• AI Fluency Framework & Foundations: (Anthropic)

• Claude 101: (Anthropic)

• Career Essentials in Generative AI: (Microsoft and LinkedIn)

• Python for Data Science: (NPTEL)

• IJRASET Certificate: (International Journal for Research in Applied Science & Engineering Technology)